Typos on a Page

Three Small Probability Models for Isolated and Bursty Errors

Applied probability note

April 23, 2026

Abstract

This note develops three elementary stochastic models for page-level typos. The first model treats pages as independent Bernoulli trials. The second introduces a latent Markov state for a fresh or tired typist, producing bursts through persistent unobserved fatigue. The third treats typos themselves as a discrete-time self-exciting point process, in the style of a Hawkes process. The aim is not to solve a high-stakes inference problem, but to show how a small change in probabilistic structure changes the qualitative behavior of a sequence of errors. Cumulative crossing probabilities are useful, but inter-typo gap distributions are the clearest diagnostic of genuine self-excitation.

1 Setup

Suppose a document is read page by page. Let

$$X_k = \begin{cases} 1, & \text{if page } k \text{ has at least one typo,} \\ 0, & \text{otherwise.} \end{cases}$$

The basic object of interest is the page sequence

$$X_1, X_2, \ldots, X_n.$$

For a threshold $q \in (0, 1)$, define the crossover page

$$n^\star(q) = \min \left\{ n : \mathbb{P}\left(\sum_{k=1}^{n} X_k \geq 1 \right) > q \right\}.$$

This quantity answers a simple question: how many pages must be read before the chance of seeing at least one typo exceeds a chosen probability level?

The more interesting question is not only when the first typo appears, but whether typos arrive as isolated accidents or in clusters. The following three models separate those ideas.

2 Model I: Independent Page Typos

The independent baseline assigns each page its own typo probability. In the original experiment this probability increases linearly with page number:

$$p_k = 0.01 + \alpha k.$$

Assuming independence across pages,

$$\mathbb{P}(X_k = 1) = p_k, \qquad X_1, \ldots, X_n \text{ independent.}$$

Then the probability of no typo in the first n pages is

$$\mathbb{P}(X_1 = 0, \ldots, X_n = 0) = \prod_{k=1}^{n} (1 - p_k),$$

so

$$\mathbb{P}\left(\sum_{k=1}^{n} X_k \geq 1\right) = 1 - \prod_{k=1}^{n} (1 - p_k).$$

This model is a clean baseline: it captures changing page-level risk but has no mechanism for clustering. Conditional on the probabilities p_k, a typo on one page carries no information about nearby pages.

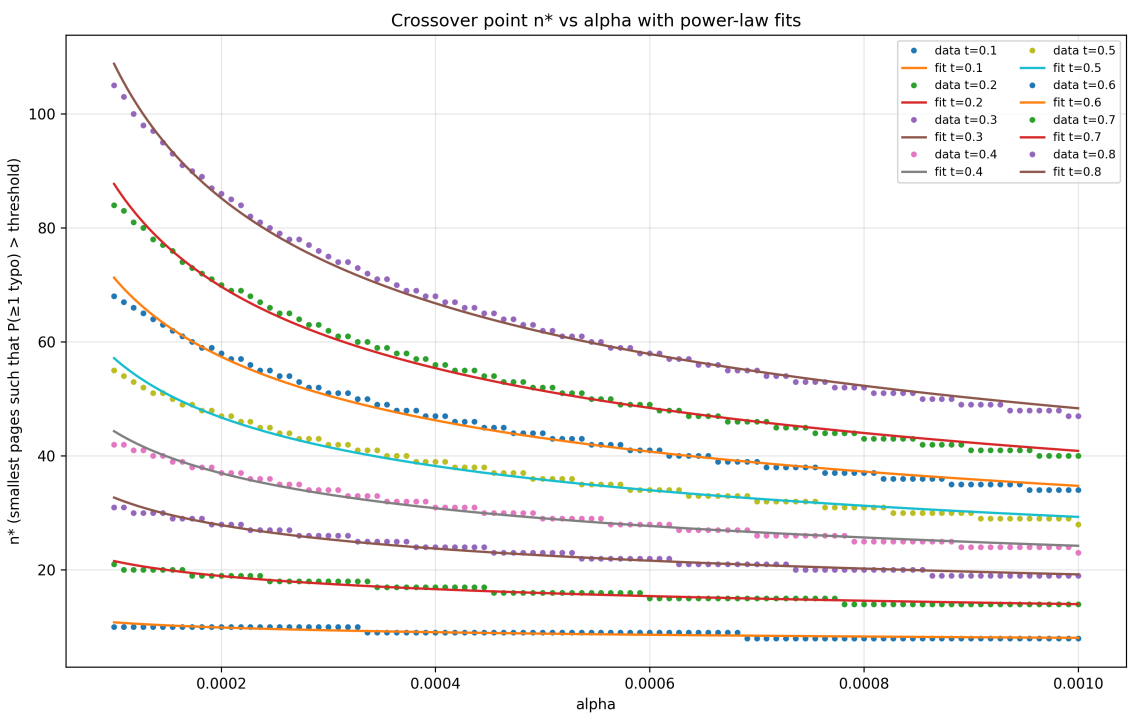

Figure 1: Independent-page model. The crossover page $n^\star$ decreases as the slope α increases. Power-law curves provide a compact empirical description of the threshold curves, although the lowest threshold is the least well described by this fit.

3 Model II: Hidden Markov Tired Typist

The second model introduces a latent state

$$S_k \in \{F, T\},$$

where F means fresh and T means tired. Typo probability depends on the hidden state:

$$\mathbb{P}(X_k = 1 \mid S_k = F) = p_F, \qquad \mathbb{P}(X_k = 1 \mid S_k = T) = p_T,$$

with $p_T > p_F$.

The state evolves as a two-state Markov chain:

$$\mathbb{P}(S_{k+1} = T \mid S_k = F) = b, \qquad \mathbb{P}(S_{k+1} = T \mid S_k = T) = a.$$

Using state order (F, T), the transition matrix is

$$P = \begin{pmatrix} 1-b & b \\ 1-a & a \end{pmatrix}.$$

The parameter a is the persistence of tiredness. The expected length of a tired run is

$$\mathbb{E}[\text{tired run}] = \frac{1}{1-a}.$$

Thus $a = 0.92$ corresponds to an expected tired run of 12.5 pages, while $a = 0.98$ corresponds to 50 pages.

3.1 Survival Recursion

The scalar product from the independent model is replaced by a two-entry survival vector:

$$v_k = \left[\mathbb{P}(X_1 = \cdots = X_k = 0, S_k = F), \mathbb{P}(X_1 = \cdots = X_k = 0, S_k = T)\right].$$

At each page, the vector is multiplied by the no-typo emission probabilities

$$d = (1 - p_F,\, 1 - p_T),$$

and then propagated through the transition matrix. In code, this is the recursion

$$v \leftarrow v \odot d, \qquad \mathbb{P}(X_1 = \cdots = X_k = 0) = v_F + v_T, \qquad v \leftarrow vP.$$

The cumulative probability of at least one typo is again one minus the survival probability.

4 Model III: Hawkes-Like Self-Exciting Typos

The third model removes the latent state and makes the observed typo process self-exciting. A typo increases the near-future typo intensity directly.

Let λ_k denote the page-time typo intensity. The discrete Hawkes-like recursion used here is

$$\lambda_k = \mu + E_k,$$

where

$$E_k = \delta E_{k-1} + \eta X_{k-1}.$$

Here μ is the baseline intensity, η is the jump in future intensity caused by a typo, and δ is the persistence of excitation. Intensity is converted to a Bernoulli probability by

$$\mathbb{P}(X_k = 1 \mid \mathcal{H}_{k-1}) = 1 - \exp(-\lambda_k),$$

where $\mathcal{H}_{k-1}$ is the history through page $k-1$.

One useful diagnostic is the approximate branching ratio

$$R = \frac{\eta\delta}{1-\delta}.$$

Values below one are the natural stable regime. The three scenarios used in the experiments are shown in Table 1.

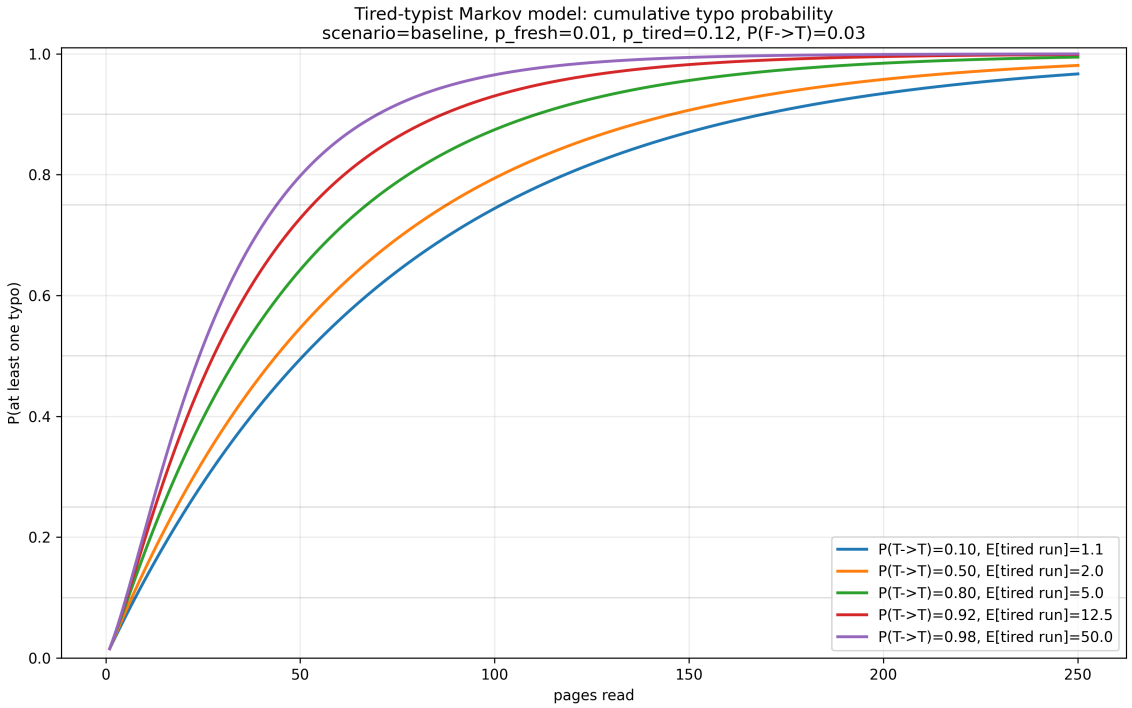

Figure 2: Hidden Markov tired-typist model. Larger tired-state persistence raises the cumulative probability of seeing at least one typo. The label E[tired run] shows the expected duration of a tired spell.

Scenario	μ	η	δ
Mild	0.012	0.10	0.45
Baseline	0.014	0.22	0.58
Volatile	0.010	0.35	0.72

Table 1: Hawkes-like page-time scenarios. The volatile case is deliberately close to the high-excitation regime, producing visibly stronger clusters.

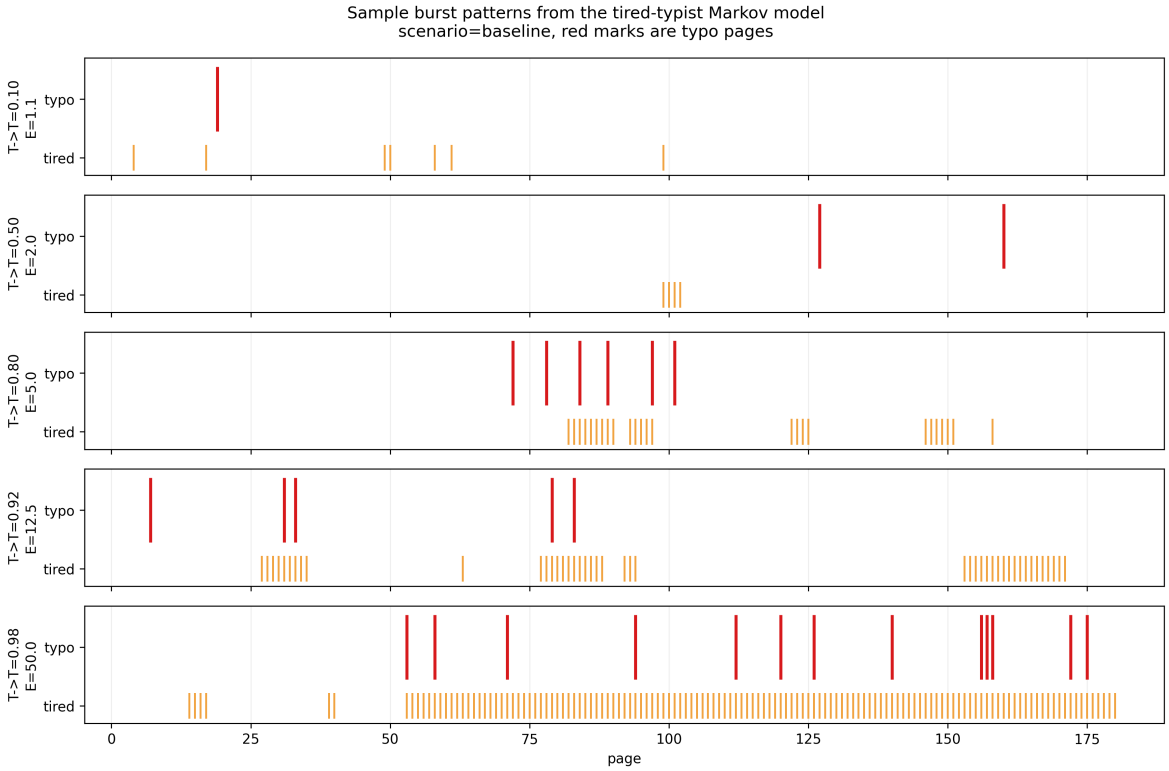

Figure 3: Sample paths from the hidden Markov model. Orange marks indicate pages where the latent typist state is tired; red marks indicate typo pages. Blank regions are fresh pages with no typo. As tiredness persistence increases, long tired stretches become visible and typos cluster within them.

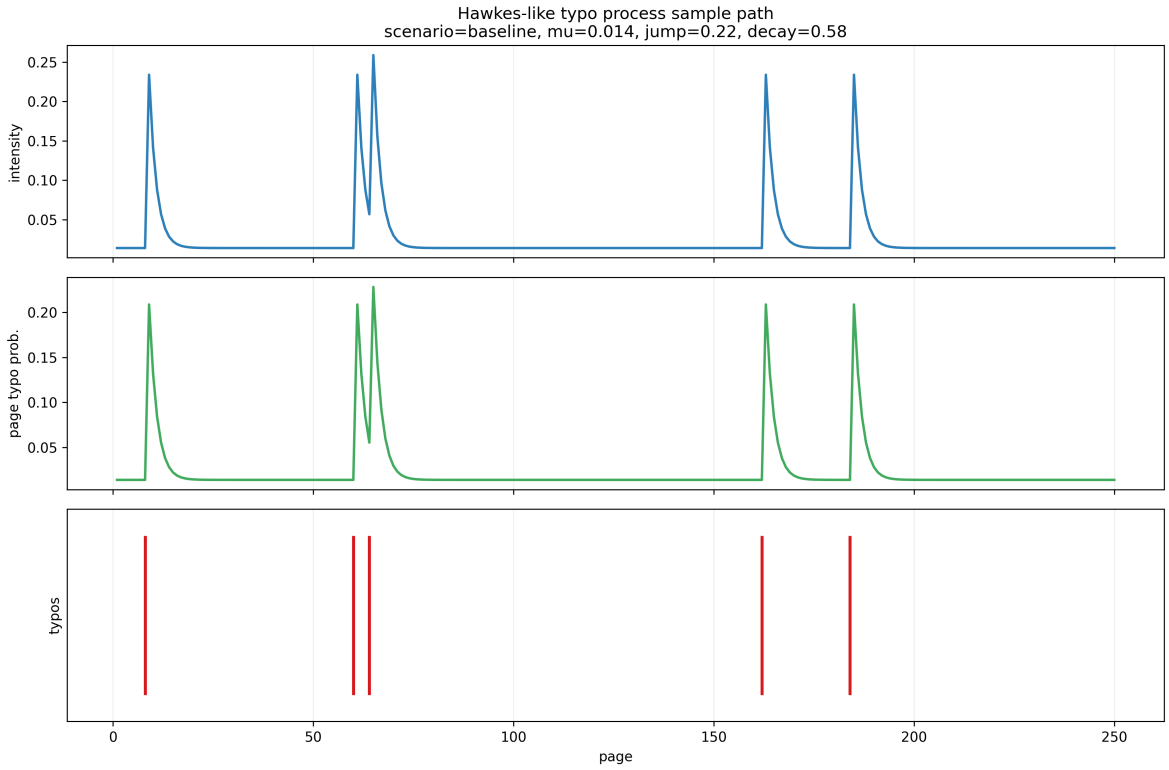

Figure 4: A baseline Hawkes-like sample path. A typo event raises the intensity and therefore the next few page-level typo probabilities. In quiet periods, excitation decays geometrically.

5 Comparing the Three Models

For the comparison plots, the independent model is calibrated to match the long-run average page typo probability of the baseline hidden Markov model. This makes it a fair non-bursty reference point. The models compared are:

1. independent calibrated Bernoulli pages;
2. hidden Markov tired typist;
3. baseline Hawkes-like self-exciting process.

5.1 Cumulative Probability

Cumulative probability curves answer when the first typo is likely to appear. They are useful, but they do not fully reveal the dependence structure. A model can have a high chance of at least one typo without being very bursty.

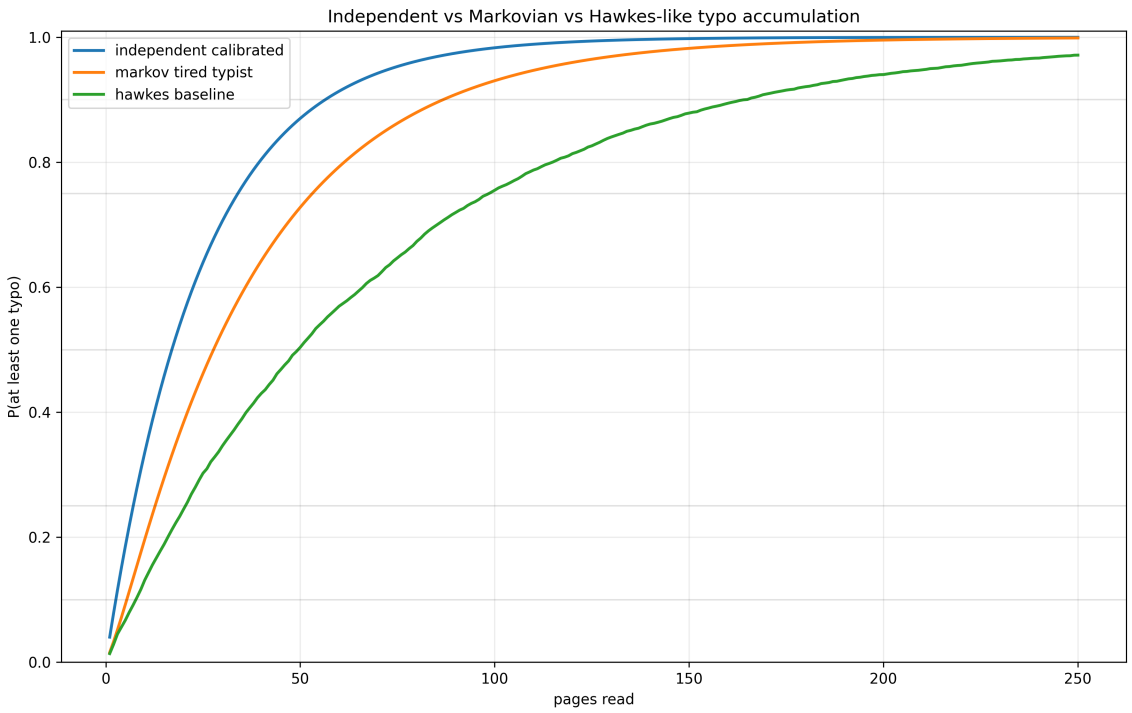

Figure 5: Cumulative probability comparison. The calibrated independent model, hidden Markov model, and Hawkes-like model can differ substantially in first crossing behavior, but this plot alone does not isolate clustering.

5.2 Inter-Typo Gaps

The inter-typo gap distribution is the sharpest illustration of self-excitation. In the independent model, gaps are roughly geometric. In the hidden Markov model, short gaps become more common because tired stretches persist. In the Hawkes-like model, a typo directly lifts the near-future hazard, placing still more mass on gaps of one or two pages.

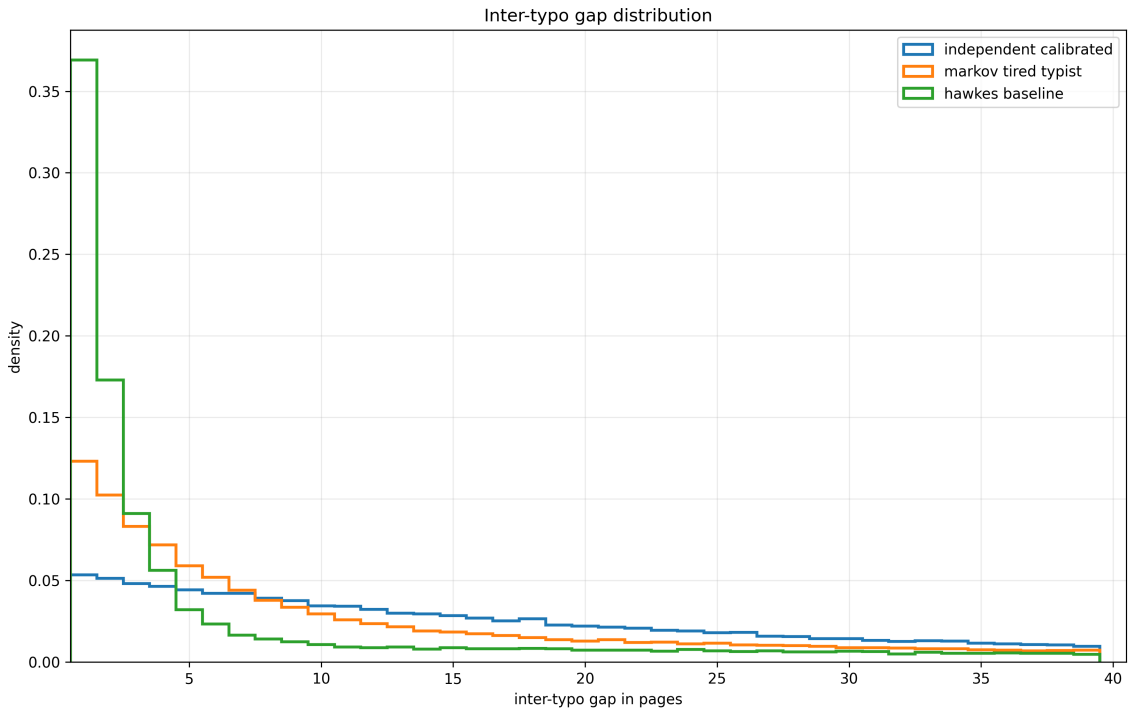

Figure 6: Inter-typo gap distributions. This is the most revealing plot in the experiment: the Hawkes-like process puts much more mass on very short gaps, showing self-excitation directly.

5.3 Burst Sizes

To summarize clusters, define a burst as a group of typos where consecutive typo pages are at most two pages apart. The resulting burst size distribution again separates isolated errors from self-excited clusters.

6 Numerical Summary

The default simulation used 8000 Monte Carlo paths of 250 pages for the simulated comparisons. The most useful summary statistic is the probability that an inter-typo gap is at most two pages:

$$\mathbb{P}(\text{gap} \leq 2).$$

In the default run, this quantity is approximately

$$\begin{aligned} \text{independent calibrated} &: 0.09, \\ \text{hidden Markov tired typist} &: 0.19, \\ \text{Hawkes baseline} &: 0.43. \end{aligned}$$

Thus the Hawkes-like model has fewer total typos than the calibrated independent and Markov baselines in this particular parameterization, but its typos are far more clustered. This is exactly the qualitative signature the point-process model was meant to expose.

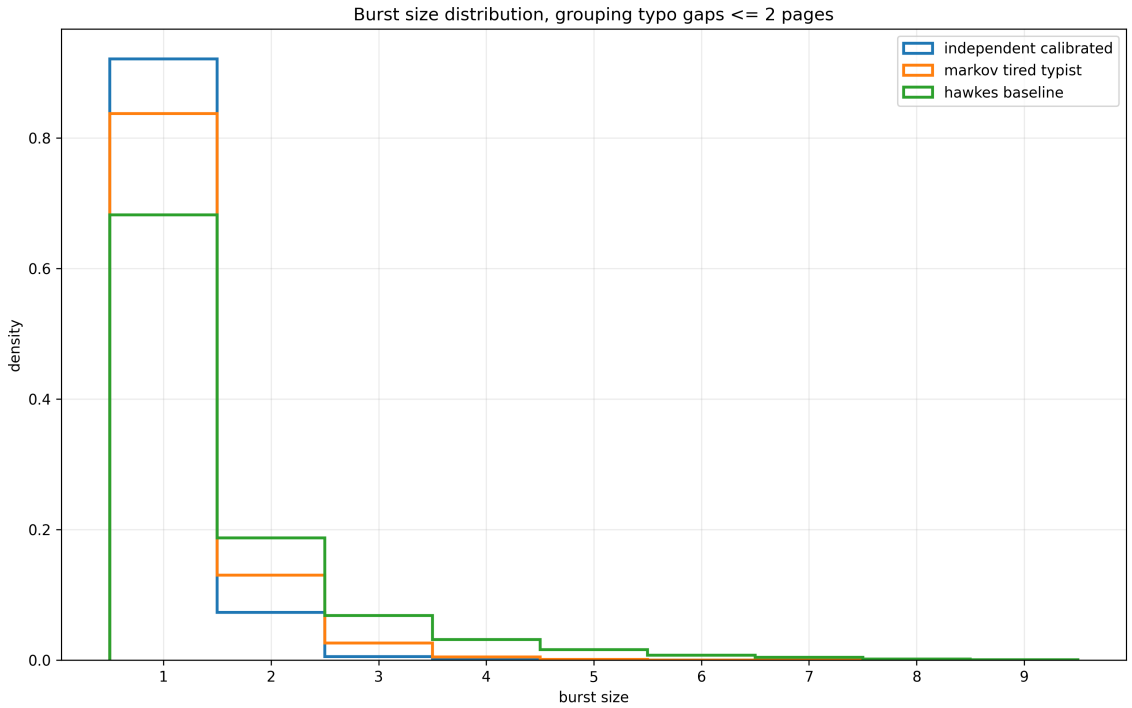

Figure 7: Burst size distributions, grouping typo gaps of at most two pages. Self-excitation increases the frequency of multi-typo bursts.

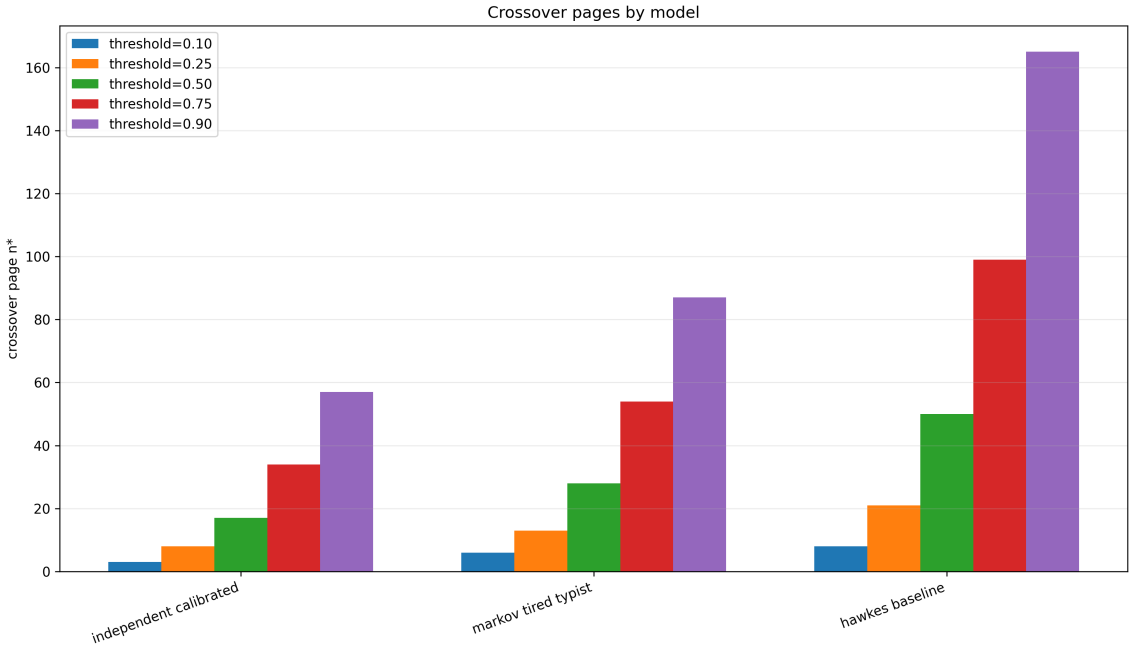

Figure 8: Crossover pages by threshold for the three comparison models. The Hawkes baseline is slower to produce the first typo in this parameterization, while still being more clustered conditional on typos appearing.

7 Discussion

The three models differ by where dependence enters.

Independent model	:	no dependence between pages,
Hidden Markov model	:	dependence through latent tiredness,
Hawkes-like model	:	dependence through observed typo events.

The hidden Markov model says that typos are evidence of an unobserved state. The Hawkes-like model says that typos themselves raise the immediate future hazard. Both create bursts, but the inter-gap distribution shows that self-excitation has a particularly direct signature: a heavy concentration of very short gaps.

The lesson is modest but useful. Even in a toy typo problem, the modeling choice determines whether errors look like isolated accidents, symptoms of a persistent latent condition, or events that excite further events. The plots make these mechanisms visible.